

Library Management System

Task 2: Normalization

1NF: Remove multi-value (unique rows)

- Split multi-valued fields into separate tables.

BookAuthors
Book_id
Author_name

BookCategories
Book_id
Category

LibraryData
member_id
member_name
book_title
borrow_date
return_date
fine_amount
publisher_name
publisher_country

Why 1NF matters:

- Removes repeating groups → easier querying

2NF: Remove **partial dependencies** (non-key attributes depending only on part of a composite key).

Actions:

- Identify composite keys (here, (member_id, book_id) in Borrow table).
- Move attributes that depend only on **member_id** or **book_id** into separate tables.

Member
member_id
member_name

Book
Book_id
book_title
publisher_id
publisher_name
publisher_country

Borrow
member_id
book_id
borrow_date
return_date
fine_amount

BookAuthors
Book_id
Author_name

BookCategories
Book_id
Category

BookAuthors & BookCategories stay the same.

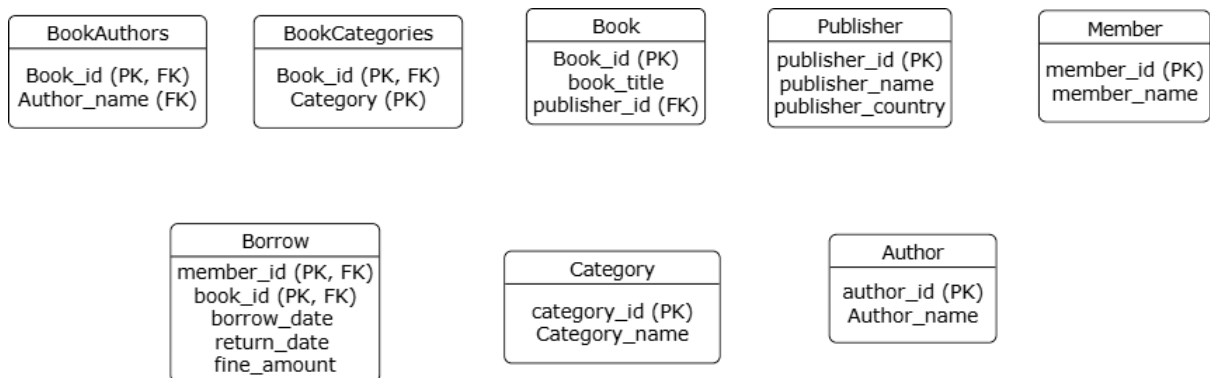
Why 2NF matters:

- Eliminates redundancy
- Updates become easier → e.g., change book title once, not in multiple rows

3NF: Remove **transitive dependencies** (non-key attributes depending on other non-key attributes).

Actions:

- publisher_country depends on publisher_name, not book_id → create **Publisher table**.
- All other attributes now depend **only on the primary key** of their table.



Why 3NF matters:

- No redundant or transitive data
- Consistency → one source of truth per attribute
- Easier maintenance and querying